

FitMenu — Intelligent Dietary Companion App

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Field of the Invention

This invention pertains to mobile software in the domain of nutrition tracking, barcode and menu scanning, and personalized diet management. It merges multiple functions into a unified platform, empowering users to make informed dietary choices in real time.

Background / Problem Statement

Individuals today face significant challenges in understanding the nutritional composition of foods they consume, especially when dining out or purchasing packaged goods. Existing solutions often specialize in a single aspect, such as barcode scanning or calorie logging, and rarely integrate restaurant menu analysis or multi-diet classification. Furthermore, most applications fail to dynamically adapt recommendations based on the user's specific health goals, dietary preferences, and restrictions. FitMenu addresses these gaps by delivering a portable, intelligent dietary assistant that works seamlessly in grocery stores, restaurants, and at home.

Target Audience

FitMenu is designed for all age groups interested in maintaining, improving, or monitoring their diet. This includes:

- Health-conscious adults seeking balanced nutrition.
- Individuals following specific diet plans such as Keto, Low-Carb, Carnivore, Calorie-counting, Mediterranean, Vegan, and others.
- People with medical dietary restrictions requiring precise nutritional monitoring.
- Fitness enthusiasts and athletes tracking macronutrient intake.
- General users who wish to make healthier choices while shopping or dining out.

Summary of the Invention

FitMenu is a comprehensive mobile application combining barcode scanning, restaurant menu lookup, and intelligent multi-diet compatibility analysis. Through a simple interface, users can scan grocery products, search or scan restaurant menus, and receive immediate nutritional details and compatibility status for their selected diet plan(s). The app personalizes all results according to user profile data such as age, weight, height, activity level, health goals, and dietary restrictions.

How the Application Works

- User creates a profile with basic details (age, sex, height, weight, activity level, dietary preferences, and goals).
- The home screen provides a daily calorie summary, quick access to scanning functions, and recommended meals.

- Barcode scanning function reads UPC codes on packaged goods and retrieves nutritional data from integrated databases.
- Restaurant menu scanning/search allows the user to search for a restaurant or scan a code/name to retrieve full menu nutrition info.
- An intelligent Dietary Compatibility Engine evaluates each item against predefined diet rules and classifies it as Allowed, Limited, or Not Recommended.
- All scanned or selected items can be logged into a daily food journal, contributing to real-time progress tracking.
- The app stores a local cache of data for offline usage and updates regularly from cloud sources.

Detailed Design Description

The FitMenu user interface is structured into five main sections accessible via a bottom navigation bar: Home, Scan, Log, Progress, and Settings. Design elements: - Color Palette: Turquoise (#1ABC9C, #16A085) as primary, Solar Plexus Gold (#FFB300) as accent, dark gray (#2C3E50) text, and light background (#F7FBFB). - Typography: Modern sans-serif fonts such as Poppins or Inter. - Home Screen: Displays calorie targets, remaining daily allowance, quick scan button, and featured healthy meal suggestions. - Scan Screen: Camera interface with real-time barcode detection, manual UPC entry option, and result display including nutritional breakdown and diet compatibility tags. - Restaurant Menu Screen: Search bar and menu list, each dish showing nutritional facts, health labels, and compatibility status for multiple diets. - Log Screen: Chronological list of meals with calorie and macronutrient summaries, plus the ability to attach images and notes. - Progress Screen: Graphs showing weight changes, daily calorie trends, and weekly summaries. - Settings: Profile editing, diet plan selection, units and language preferences, privacy controls, and integration with Google Fit/Apple Health.

Claims

- A mobile application integrating barcode scanning, restaurant menu data retrieval, and a multi-diet compatibility engine in a unified platform.
- A method of evaluating food items and meals for compatibility with various diets based on real-time data from multiple integrated sources.
- A personalized dietary recommendation system adapting to user-specific health goals and restrictions, with both online and offline functionality.

Advantages of the Invention

- Combines grocery and restaurant nutritional data into one streamlined platform.
- Provides immediate, diet-specific compatibility feedback for any scanned or searched food item.
- Works both online and offline with regularly updated data caches.
- Personalizes all recommendations based on user goals, restrictions, and preferences.
- Supports a wide range of dietary plans simultaneously.

Potential Applications

- Daily nutritional tracking for weight management and health maintenance.
- Guidance for individuals on specific medical diets requiring strict nutrient control.
- An educational tool for nutritionists, coaches, and healthcare providers.
- Integration with wearable devices for comprehensive health monitoring.

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